RESPONSE TO THE REVIEWERS

REVIEWER 1

Reviewer Point P 1.1 — Jablonka et al give a sweeping, well-structured review on the promise of general-purpose models for the chemical sciences. I really like the balance of conceptual clarity and technical depth throughout the article, particularly in the figures that illustrate model workflows and representative examples. In my opinion, it does a great job covering multiple aspects of the GPM pipeline and I think this will be a nice contribution to the field. Overall I enjoy reading it and think it can be accepted with some minor revisions.

Reply: *We sincerely appreciate the reviewer's positive feedback on our manuscript's contribution.*

Reviewer Point P 1.2 — The article swings between LLMs and broader GPMs a bit inconsistently. Sometimes it reads as if the topic is "large language models for chemistry", and other times it zooms out to cover the broader general-purpose model landscape, including other architectures. I think the introduction could be clearer in stating exactly what is being covered, if over 80% is for LLMs (or transformer-based foundation models), I think some modifications may be needed for title and abstract. Otherwise I feel somehow hard to get a sense of the entire field of GPM in Chemistry (and indeed this is a big scope). A one-sentence thesis and consistent reminders throughout the review would help readers track the core argument.

Reply: *To provide clearer direction, we have now introduced a new table (see Table R1) early in the paper that explicitly defines s (GPMs) and distinguishes them from other model types (domain-specific foundation models and specialized chemistry pipelines). This table provides concrete examples and clearly contrasts GPMs with domain-specific foundation models (e.g., AlphaFold,[1] ESM,[2] MACE-MP-0[3]) and specialized chemistry pipelines (e.g., QSPR models, graph-based reaction predictors).*

Table R1: Illustrative examples of GPMs, domain-specific foundation models, and specialized chemistry ML pipelines. The table depicts the definition of GPM with examples for such models, as well as comparisons with domain-specific models and chemistry pipelines. Note that a GPM does not necessarily output text.

Category	Typical Characteristics	Representative Examples
GPMs	Pre-trained on a large heterogeneous corpus spanning multiple modalities. Supports zero/few-shot generalization and can be fine-tuned for diverse chemistry tasks. Capable of autonomous agent behavior, including planning and execution.	**Autoregressive:** GPT-4,[4] LLaMA,[5] Galactica[6] **Diffusion-based:** Gemini Diffusion,[7] Inception Mercury[8] **Other:** Mamba-based[9] models
Domain-Specific Foundation Models	Trained on curated, domain-specific datasets (e.g., protein structures, crystal structures). Achieve state-of-the-art performance in narrow task sets, but are typically not multimodal or generalizable to unrelated chemistry problems.	AlphaFold,[1] ESM,[2] MACE-MP-0,[3] MatterSim,[10] MolecularTransformer[11]
Specialized Chemistry Pipelines	Domain-specific models combined with rule-based or symbolic components. Often rely on hand-crafted descriptors with limited transferability beyond the target task.	Graph-based reaction outcome predictors, QSPR models using Morgan fingerprints, GPR on NMR shifts

Additionally, we added the following text in section 1, and we reviewed the usage of GPM and large language model (LLM) throughout the manuscript to ensure that they are correctly applied according to this definition.

> *A GPM is a model that has been pre-trained on a broad, heterogeneous corpus spanning multiple data modalities (text, images, graphs) or representations (e.g., common names, 3D coordinates, molecular images). It can be applied to a wide spectrum of downstream tasks that differ in objective (classification, regression, generation, reasoning), input format, and domain—ranging from natural-language processing to chemistry and vision—with little or no task-specific fine-tuning. A GPM supports zero-shot, few-shot, or transfer learning and can serve as the core component of autonomous agents.*

In addition, we also updated the title of the manuscript to now read "General-Purpose Models for the Chemical Sciences: LLMs and Beyond".

Reviewer Point P 1.3 — Along those lines, Section 2 and 3 are strong technically, but could benefit from reiterating why these details matter specifically for chemistry-focused GPMs (instead of looking like a general background section for ML in Chemistry review paper). There's a lot on tokenization, embeddings, and architectures (which I appreciate), but the chemical intuition sometimes gets lost. While figures do include some chemistry example, some synthetic chemists without too much background may find confusing. I'd suggest adding short framing sentences to keep readers anchored in chemistry even as the model mechanics get dense.

Reply: *To address this concern, we have now integrated "example boxes" throughout our article, especially in section 3. We hope these dedicated boxes walk readers through key* *concepts with chemistry-specific examples, helping to maintain the connection to chemistry even as we discuss technical details about models' architectures. Below is one such example box we added to illustrate agent-based reasoning for chemical structure elucidation.*

Agent-based reasoning for chemical structure elucidation

User Query: *"Identify this unknown white solid organic compound from analytical data."*
Agent Reasoning Cycle:

1. **THINK:** *"Unknown white solid. Need to analyze all spectra systematically."*
2. **ACT:** *Analyze `IR` spectrum tool → Output peak data (3000 cm^{-1} to 3500 cm^{-1} broad, 1680 cm^{-1} sharp, 1600 cm^{-1}, 1500 cm^{-1}).*
3. **ACT:** *Analyze `MS` tool → Output peak data (`Top 3 peaks (135.0, 26.48), (106.9, 0.43), (120.4, 0.31)`).*
4. **OBSERVE:** *"Broad `O-H` stretch suggests carboxylic acid. Sharp `C=O` at `1680` indicates aromatic carboxylic acid. Aromatic `C=C` stretches confirm benzene ring, and `MS` tool suggests molecular formula $C_8H_8O_2$."*
5. **THINK:** *"Aromatic carboxylic acid with $C_8H_8O_2$... probably substituted benzoic acid. Need to determine substitution pattern."*
6. **ACT:** *Run ^{1}H `NMR` integration tool → Input chemical shifts (`7.8 ppm` doublet, `7.3 ppm` doublet, `2.4 ppm` singlet, `12.1 ppm` broad).*
7. **OBSERVE:** *"Two aromatic doublets suggest para-disubstitution. Singlet at `2.4 ppm` integrates for `3H` — likely CH_3. Broad peak at `12.1 ppm` confirms COOH."*
8. **THINK:** *"Para-disubstituted benzene with CH_3 and COOH groups. Could be p-toluic acid."*
9. **ACT:** *Search chemical database for "`para-methylbenzoic acid`".*
10. **OBSERVE:** *"`4-methylbenzoic acid` (p-toluic acid) matches: $C_8H_8O_2$, `mp` 180 °C."*
11. **ACT:** *Run structure validation tool → Compare predicted vs. observed spectra.*
12. **OBSERVE:** *"Predicted ^{1}H `NMR`: δ `7.9 (d, 2H), 7.2 (d, 2H), 2.4 (s, 3H), 12.0 (br, 1H)`. Perfect match!"*
13. **ACT:** *Cross-reference `MS` fragmentation pattern tool.*
14. **OBSERVE:** *"Base peak at `m/z 91` (tropylium ion, loss of COOH), molecular ion at `136`. Consistent with p-toluic acid fragmentation."*

RESULT: *Structure confirmed as `4-methylbenzoic acid`. All spectroscopic evidence supports this assignment.*
The agent systematically combines `IR`, `NMR`, and `MS` data with database searching to solve a realistic structure determination challenge.

Reviewer Point P 1.4 — Section 5.2, which lists existing GPMs for chemical sciences, feels very thin compared to the rest. This is a missed opportunity to give readers a map of the landscape. Many of the examples are mentioned quickly but not deeply described. I think most readers would want to know: which model should I try and why?

Reply: *Following this suggestion, we have now expanded Section 5.2 to provide a more comprehensive map of the GPM landscape for chemistry. We added a detailed table (see Table R2) summarizing the key characteristics of available models—including their architecture (e.g., encoder-only, decoder-only, encoder-decoder, base model), training approach, representative tasks, and target systems (organic molecules, inorganic materials, proteins, etc.). This table helps readers understand the landscape of available GPMs and make informed decisions about which model to use for their specific applications.*

Table R2: Non-comprehensive list of existing GPMs in chemical sciences. Most current models are trained on a mixture of general and domain-specific corpora. Models are sorted by the order they appear in the main text.

Model	Architecture	Training	Representative Tasks	Representative Systems
DARWIN 1.5[12]	decoder-only transformer (LLaMA) with 7B parameters	fine-tuning on Q&As; multi-task fine-tuning on language-interfaced classification and regression tasks	materials property prediction from natural language prompts	inorganic materials
ChemLLM[13]	decoder-only transformer (InternLM2) with 7B parameters	instruction-tuning on template-based qa using SFT or DPO	multi-topic chemistry Q&A	organic molecules & reactions
nacho[14]	encoder–decoder transformer (T5) with 250M or 780M parameters	pre-trained using SSL on scientific literature and SMILES strings; instruction-tuned	natural language to SMILES; small-molecule property prediction	organic molecules & reactions
LLaMat[15]	decoder-only transformer (LLaMA) with 7B parameters	continued pre-training on materials literature and crystallographic data	materials information extraction; understanding and generating CIFs	inorganic materials
ChemDFM[16]	decoder-only transformer (LLaMA) with 13B parameters	pre-trained on 34B tokens from chemistry papers and textbooks; instruction-tuned with Q&A and SMILES	SMILES to text; molecule/reaction property prediction	organic molecules & reactions
ethero[17]	decoder-only transformer (Mistral-Small) with 24B parameters; MoE	SFT on reasoning traces, then RL	molecular editing; one-step retrosynthesis; reaction prediction	organic molecules & reactions

Reviewer Point P 1.5 — Section 5 and 6 make some strong claims about LLM-assisted discovery and autonomous science. But in practice, one risk can be most of these systems haven't gone beyond cherry-picking or recombining known ideas. One interesting thing author can consider (optional, totally up to them) is try to include a few concrete "failure cases" or borderline examples where the model output seems good on the surface but breaks under scrutiny. This could be framed as "cautionary tales" rather than critiques, which would add balance and practical value.

Reply: *We have now reviewed Sections 5 and 6, moderating the tone of the claims we make and adding explicit **Limitations** and **Open Challenges** subsections in each application section. For example in hypothesis generation section (section 5.4) we added*

> **LIMITATIONS** *Current LLM-driven systems lack the kind of creativity needed for paradigm-shifting hypotheses, tending to rearrange training data and retrieved content rather than propose genuinely new mechanisms. As a result, the evaluation of such outputs is also fragile, because using proxy metrics for "novelty" or "impact" that poorly track real scientific value can be deceiving. Finally, the problem's open-ended nature makes systematic benchmarking ill-posed.*

In addition, we added example boxes that focus and conceptualize some failure cases (see response Point P 1.6)

Reviewer Point P 1.6 — Section 5 on planning and executing experiments are exciting, but a bit abstract. Are there concrete experimental workflows, for example, I know in past a few years some work in LLM + porous materials/polymer/ nanoparticle/ electrochemistry, that have been executed in real labs using these GPMs? If yes, even one well-described example would help readers from the experimental side understand what's actually possible. Also I think many examples the author mentioned are more on "digital discovery" side where the LLM were tested on real experimental dataset but not necessarily the verification are performed in real lab. So including some work with real lab verification (new discovery? not sure) may make a good balance and look convincing to experimentalist readers.

Reply: *We have now added an illustrative example as a new figure (see Figure R1) in the planning section, demonstrating how an LLM can plan chemically to identify pairs of inorganic salts in solutions by decomposing the problem into sequences of experiments (acid/base tests). Importantly, this figure also highlights the limitations of current LLM planning capabilities by showing problematic steps in the plan with linked annotations explaining the errors.*

Additionally, in the experiment execution section, we now added concrete real-world examples (from ChemCrow and ORGANA) shown below,

Example of LLM for chemical planning

Prompt: You are given three different solutions (A, B, and C), each containing a different pair of inorganic salts from the following list, but you do not know which pair of salts are present in each solution: KCl, $AgNO_3$, $Pb(NO_3)_2$, BaI_2, Na_2SO_4, $Ni(ClO_4)_2$ in addition to the three unknown solutions, you have access to 1M KOH and 1M HNO_3 and no other solution. You can mix the unknown solutions together or add acid or base to them. Your task is to identify which pair of salts is present in each solution. Decompose this problem into a sequence of experiments. Explain which tests to perform, how to measure outcomes, and how to iteratively refine the hypotheses to reach the final conclusion.

Answer:
Step 1: Analyze tools and constrains we have observation types: ......
Solubility rules:
- Ag^+: insoluble with Cl^- (AgCl ppt), soluble otherwise
- Ba^{2+}: insoluble with SO_4^{2-} ($BaSO_4$ ppt)

......

Step 2: Individual base test (add KOH) ......
Identify Ag^+ (brown/black), Pb^{2+} (white), Ni^{2+} (green) by ppt color

Step 3: Identify halides and sulfates via mixing
Mix unknown solutions in pairs (A+B, A+C, B+C). Observe halide/sulfate precipitation:
AgCl (white ppt), PbI_2 (yellow ppt), $PbSO_4$ (white ppt), $BaSO_4$ (ppt).

Step 4: Dissolve ppts with HNO_3
Ag_2O, $Pb(OH)_2$, $Ni(OH)_2$ dissolve in HNO_3
-Take ppt formed in step 2
- If ppt dissolves: confirms the cation identify

Step 5:Iteratively refine assignment
Suppose after KOH addition:
A → green ppt → contains Ni^{2+}
B → brown ppt → contains Ag^+
.....
Then mix solutions:
B + A → white ppt → indicates Ag^+ + Cl^-
→ solution B has Ag^+, A has Cl^-
......
Combine observation:
A → Ni^{2+} + Cl^-?
B → Ag^+ + I^-?
C → Pb^{2+} + SO_4^{2-}?

Step 6: Cross-check final assignments
Mix assigned solutions → no new ppt unless expected.
Dissolve expected hydroxides with HNO_3 → confirms cation identity.
Double-check unusual cations like K^+ and Na^+ (remain in solution, soluble).

Incomplete domain knowledge: Omits Ag_2SO_4, AgI precipitation

Redundant step: Unnecessary confirmation steps with no new information

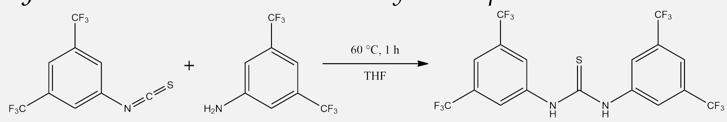
Systematic incompleteness: Only partial ions are recognized, not two complete salts in one solution

Chemical error: Ag^+ and I^- cannot coexist in same solution (B), Pb^{2+} and SO_4^{2-} cannot coexist in same solution (C)

Figure R1: An example of using LLM for chemical planning to decompose problem. LLM decomposes a chemical problem into sequential steps with detailed procedures. We manually evaluate the plan step by step, and highlight the problematic steps in text boxes, each linked to the corresponding reason. Plan generated by OpenAI GPT-5 (ChatGPT, web).

Planning and executing experiments with GPMs

ChemCrow: *When prompted with the task "Find and synthesize a thiourea organocatalyst which accelerates a Diels-Alder reaction", ChemCrow sequentially queried tools to gather information and solve the task:*

> *"First, I need to find a thiourea organocatalyst that can accelerate the Diels-Alder reaction. I will perform a websearch to find a suitable catalyst.*
> ***WebSearch tool:*** *Schreiner's thiourea catalyst*
> *Now I will obtain the SMILES.*
> ***Name2Smiles tool:***
> `FC(F)(F)c1cc(CN(=S)Nc2cc(C(F)(F)F)cc(C(F)(F)F)c2)cc(C(F)(F)F)c1`
> *I will plan a synthesis for Schreiner's thiourea catalyst.*
> ***SynthesisPlanner tool:*** *Detailed synthesis plan*
>
> 
>
>
> *I will execute the synthesis.*
> ***SynthesisExecuter tool:*** *Successful synthesis."*

The agent proposed to synthesize the target molecule from the corresponding isothiocyanate and aniline derivatives and executed the synthesis on the RoboRXN platform.[18] *The product was confirmed using HPLC/MS:* `m/z 501 [M+H]` *calculated, found* `m/z 501.02`*.*

ORGANA*: ORGANA interacts with chemists using LLMs to derive experiment goals and makes decisions with chemists in the loop to control robots and lab devices.*

- ***Start-up:*** *Users give their intention "I want to create a Pourbaix plot of a quinone, Sodium Anthraquinone-1-sulfonate.", and start interact with ORGANA:*

  ORGANA: What is the setup of the lab environment? What hardware and reagents are present?

  User: I have a 2 mM *solution of the quinone......*

  ORGANA: Can you provide an example of how to run the experiment?

  User: Add 2 mL *of quinone......*

  ORGANA: What is the expected outcome of the example experiment?

  User: The pH of the solution is expected ......

  ORGANA: How many experiments are we planning to run in total?

  User: 6

  ORGANA: Ok, thank you! I will start the experiment......

Planning and executing experiments with GPMs (ctd.)

- ***Ambiguity and Uncertainty Mitigation:*** *ORGANA.Reasoner evaluates whether experiment outcomes align with user expectations. If it can't interpret the results, it alerts the user to check for environmental issues. The user provides feedback explaining the outcomes, which ORGANA.Reasoner then uses to refine its next plan:*

  "I'm not sure what happened, but repeat the previous experiment just in case"

  "Nothing is wrong, carry on"

  "The pump was stuck, it's ok again"

- ***Experimental Reports:*** *ORGANA reports the experimental logs and summaries:*

  "A series of experiments was conducted to measure the potential of a quinone solution at various pH levels, specifically from pH7 *to* pH9*...... After performing the experiments, these are the results: The estimate for* pK_{a1} *is* 8.096*. The estimate for* pK_{a2} *is* 12.380*. The estimate for slope is* −60.958*."*

- ***Results Comparison with Chemists:*** *Comparison of Pourbaix diagrams and first-region slope estimates from electrochemical experiments, with ORGANA using three measurement points per pH region and chemists using four. ORGANA yields results comparable to those of chemists: for* pK_{a1}*, ORGANA produces* 8.03 *and chemists* 8.02 *and for the estimated slope, ORGANA obtained* −61.3 mV/pH *while chemists obtained* −62.7 mV/pH.

Reviewer Point P 1.7 — The limitations sections (in 5 and 6) spread out in multiple parts of the paper and many could be more comprehensive. A key concern for many chemists is that LLMs often produce plausible but wrong predictions (not necessarily just hallucinations), especially for numerical tasks (but not the case for GPM, since language model's tokenization cause the trouble). The idea that LLMs can predict numerical properties sounds attractive, but when tested rigorously, their performance may be worse than baseline models trained on just a few hundred datapoints, but if only compared with some bad baseline they still look okay. In some other cases, data leakage already happen during the training, so for some more trivial chemical task they can easily win because answers were memorized. Some examples author provide fall into above category (even those work published in peer review journal) I'd suggest making this tension more critical and explicit: what's the tradeoff between flexibility and accuracy? What should and should not be done?

Reply:

In the initial version, we highlighted through explicit meta-analysis the performance comparison across different model architectures. For example, Figure R3 compares normalized error distributions for GNN-based models, LLM approaches, and baseline methods across materials property prediction tasks, clearly demonstrating where LLMs underperform compared to specialized models.

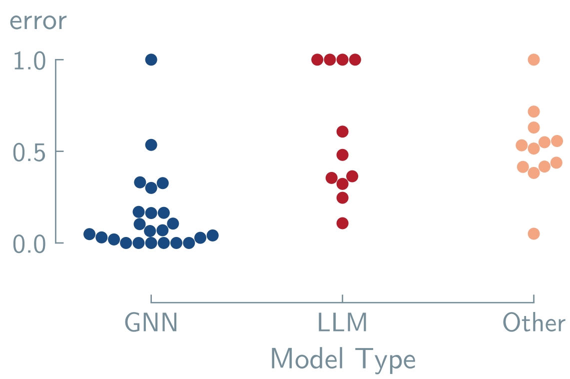


Figure R2: Normalized error distributions for materials property prediction models across different architectures. Each point represents the normalized error of a model on a specific property prediction task. Normalization was achieved with min/max values of each dataset to produce a range of errors between 0 and 1. The first column (blue) shows graph neural network (GNN) based models, the second column (red) displays LLM approaches, and the third column (orange) represents other baseline methods and state-of-the-art (SOTA) models, including *CrabNet*.[19] Lower values indicate better predictive performance. Data adapted from Alampara et al. [20]

In the revised version, we have now additionally consolidated and carefully added known ***Limitations*** *and* ***Open Challenges*** *into separate subsections in each of our application sections. These sections are intended to make readers aware that, despite the existing work, GPMs, and especially LLMs, still face significant problems for many tasks, such as property prediction. For example, in the section of the figure above (section 6.1) we added the following limitations and open challenges section.*

> **LIMITATIONS** *A significant challenge for GPMs in chemistry lies in managing dataset limitations and selecting appropriate chemical representations. Practical applications often suffer from highly unbalanced datasets, where examples of optimal materials are vastly outnumbered by poor-performing ones, forcing difficult compromises that can diminish model performance.[21] The choice of how a material is represented (SMILES notation versus International Union of Pure and Applied Chemistry (IUPAC)) also critically impacts performance, indicating that data preprocessing remains a crucial consideration. Beyond data issues, architectural constraints present another barrier as illustrated in Figure R3. Alampara et al. [20] reveal that while LLMs are effective for tasks relying on compositional information, they struggle to interpret geometric or spatial data when it is encoded in text. This suggests a fundamental limitation of transformer-based architectures for applications requiring spatial reasoning. Consequently, the conventional assumption that performance can be universally improved by scaling up model size or pre-training data is challenged.[22] Such scaling may not overcome the inherent bias against geometric understanding.*
>
> **OPEN CHALLENGES**
>
> - ***Encoding 3D Structure***: *Developing methods to effectively represent and integrate geometric, spatial, and structural information to overcome their inherent bias toward compositional data.*
> - ***Dynamic Knowledge Integration***: *Move beyond static fine-tuning to establish a reliable, real-time framework that can incorporate new, evolving scientific findings without catastrophic forgetting.*

- ***Fundamental Architectural Shifts**: Exploring whether many of the existing GPM architectures are sufficient or if new, specialized ones are needed to capture complex relationships in molecular and materials science.*

Reviewer Point P 1.8 — Related to above, there is little mention of how LLMs might be misled—by improper RAG, low-quality fine-tuning datasets, or naively designed tasks. E.g. user looking for a material, RAG gives an out-dated or retracted article and LLM becomes confident in its answer it because the way RAG works. or user intentionally prompts or fine-tune in a way the output can be controlled so they end up with good looking scores. Some discussion around the risks. Are there best practices for validation when using these models in the wild? This would help readers adopt the tools responsibly.

Reply: *Following this suggestion, we now discuss the problems with RAG in Section 5.3.3, where we state in the limitations section:*

> *LLMs and LLM-based systems are valuable information-gathering tools, but they have critical limitations. Their knowledge becomes outdated immediately after training. Without additional tools, they cannot identify when retrieved sources have been retracted or corrected.*
>
> *Effective knowledge gathering requires human-artificial intelligence (AI) collaboration. Current systems struggle to ask relevant clarifying questions.[23] Domain-specific benchmarks for evaluating this capability remain absent, hindering progress in developing truly interactive knowledge-gathering agents.*

and in the 5.3.4 section, the open challenges read

- ***Multimodal Understanding.** Current models cannot reliably extract data from plots, tables, and figures.[24,25] This severely limits data extraction because substantial chemical knowledge exists only in visual formats. Improved multimodal capabilities are essential for comprehensive literature mining.*
- ***Source Quality Assessment.** Citation counts and journal impact factors provide crude quality signals. How to select high-quality sources. Scholarly metrics alone are insufficient.*

This discussion addresses the risks of using RAG with outdated or retracted sources, as well as other potential failure modes.

Towards the general best practices for validation, we have now added an example box summarizing some of the principles in the design of the evaluation section (section 4.2), see below:

Best practices for robust model evaluation

Setup and Reporting

- *Ensure full reproducibility: document all steps from data preparation to scoring.*
- *Fix and report: model checkpoint or application programming interface (API) version, date and time, inference settings (e.g., temperature, context length).*
- *Store all evaluation assets (prompts, datasets, scripts) under version control for traceability.*
- *Clearly define: evaluation goal, task scope, data origin, and underlying assumptions.*

Validation and Sanity Checks

- *Check for potential training data leakage by testing partial completions or measuring accuracy drop under paraphrasing.*[26]
- *Use clear-cut factual questions phrased in multiple ways to test consistency.*
- *Include prompts containing obvious errors (e.g., "benzene is not aromatic") to probe factual robustness.*
- *Perform repeated runs with different random seeds or sampling temperatures to estimate uncertainty.*

Bias and Construct Validity

- *Verify that tasks measure intended skills (reasoning vs. recall).*
- *Balance question difficulty; include control and counterfactual items.*
- *Check dataset composition for chemical space or label bias.*

Publication and Communication

- *Double-check factual claims against trusted references.*
- *Report both metrics and qualitative error patterns.*
- *Use structured evaluation cards*[27] *for transparency.*

Reviewer Point P 1.9 — A controversial topic: how much do GPMs really understand chemistry? Do they need to understand chemistry in order to be good at outputting chemistry answers? If yes, how can they be improved? If no, how to learn real insights from benchmarking?

Reply: *We have now added some discussion about it in Section 8 (Outlook and Conclusions). Which now reads,*

> *It remains unclear whether generative models achieve a genuine understanding of chemistry or merely excel at pattern recognition, a distinction obscured by the lack of methods to quantify chemical reasoning.*[24] *This uncertainty challenges the need for human-interpretable output, implying that the latent knowledge within hidden embeddings may be more significant.*[28]

Additionally, we also added a paragraph in the same section discussing when to use a GPM. It reads:

> *With all this flexibility, it becomes tempting to deploy GPMs across a wide range of problems in the chemical sciences. However, as discussed in the previous*

> *sections, they are not always a direct replacement for existing approaches. In some cases, using a GPM may be unnecessary and inefficient. That said, there are clear scenarios where GPMs offers distinct advantages. When data is unstructured or "fuzzy"—such as text-based descriptions or incomplete experimental records lacking explicit molecular structures—GPMs, become valuable alternatives where specialized pipelines like GNNs would be inapplicable. Similarly, in extremely low-data regimes where specialized pipelines or domain-specific foundation models would overfit, GPM embeddings can provide more robust predictions than random baselines. Finally, for dynamic environments that require iterative reasoning, interpretation, and adjustment of the system states, GPM-powered agents are well-suited. In contrast, when clean datasets are available and inductive biases are well-understood, domain-specific models typically offer superior performance and interpretability.*

We would also like to highlight that we had some meta-analysis showing the performance comparison across different model architectures. For example, Figure R3

Reviewer Point P 1.10 — One area that I think deserves more attention is databases. Section 2 and Table 6 touch on data creation, but for foundation models the quality and diversity of pretraining data is central. Are there enough large-scale, clean chemical datasets for pre/post training? What about coverage of underexplored materials or experimental data? What percentage are simulated data / measurement data? This is a practical bottleneck that's worth calling out more directly.

Reply:
We have now added some general discussion at the end of Section 2 about the lack of data quantitatively, but also qualitatively (e.g., negative experiments data is still missing) to illustrate that the available data to train such models is orders of magnitude smaller than the ideal amount of data needed and that it is already out there for other scientific domains such as math. The text now states:

> *Future Directions*
>
> *A primary obstacle in the development of GPMs for chemistry is the immense scale of data required for pre-training, which reaches into the trillions of tokens. This demand is illustrated by models like Llama 3, trained on 15 trillion tokens. Yet the largest open-source chemistry corpus available contains only approximately 75 billion tokens.[29] Beyond its insufficient volume, this dataset is constrained by restrictive licenses and is not ideally suited for the primary pre-training phase. Furthermore, existing data resources lack documentation of negative or failed experiments and reasoning data related to routine laboratory tasks. The absence of such data impedes the development of robust chemistry problem-solving and planning capabilities in GPMs. This situation stands in contrast to fields like mathematics, where initiatives such as DeepSeek have successfully leveraged large, domain-specific datasets—for instance, 120 billion math tokens—for continual pre-training.[30]*
>
> *Despite the apparent difficulty of amassing diverse data on this scale, we contend that this challenge is accessible through a coordinated community effort.*

*In addition, for some application sections where data is a limiting factor, we added comments in the new **Limitations** and **Open Challenges** sections. For instance, in the Retrosynthesis section, we now include in the limitations:*

> *GPMs inherit the fundamental constraints associated with domain-specific models for retrosynthesis, which primarily concern data and evaluation methodologies. The prevailing reliance on patent-derived data introduces a significant bias, as these datasets are dominated by a limited set of reaction types and predominantly feature single-step examples. Furthermore, the absence of negative results and the*

> *frequent omission of experimental conditions can mislead model training.*[31–35] *The evaluation paradigms suffer from a parallel shortcoming, as they are largely designed for single-step routes and fail to adequately capture the complexities of real-world, multi-step case studies.*[36–38]
>
> *A further significant challenge, exacerbated in the context of GPMs, is data leakage. The vast scale of data used for training these models often leads to saturated benchmarks, where high performance may reflect memorization rather than genuine generalization.*

Similarly, one of the open Challenges of this section states:

> ***Data Quality and Bias**: To build more robust and generalizable models, the field must integrate higher-quality data from peer-reviewed scientific literature. This requires the development of standardized extraction schemas and data cleaning protocols to create a balanced and reliable training corpus.*[39,40]

Reviewer Point P 1.11 — The figures are visually appealing and help organize the concepts. That said, some are too generic. Many look like a perspective-style sketch rather than a reference figure. Consider adding more concrete visual examples of inputs, outputs, or workflows from real models—especially for readers newer to GPMs. I like some figures do give some simple examples that can be very helpful, but some others seem to be beneficial from more details than just some icons.

Reply: *We have now reviewed the figures, removing some of them that we considered non-helpful in Sections 5.8 and 7.1 (e.g., old Figure 23 in the education section, which was icon-heavy and did not provide additional information). In addition, we added a figure with a more specific chemical example in Section 5.5 to provide readers with more concrete visual examples of inputs, outputs, and workflows.*

Reviewer Point P 1.12 — I also think a short paragraph somewhere (Section 7 probably?) on the resource cost of GPMs would be a good addition. Fine-tuning or even training large parameter model isn't trivial. For groups used to training conventional ML models or doing DFT, the compute and environmental footprint of these models may be surprising. A short comparison table or rough estimate would be useful for context.

Reply:
We have now added the cost in terms of both time, data and energy in the model adaptation table in section 3.11 (see Table R3)

Table R3: Model Adaptation Approaches Overview: This table provides a rough overview of the estimated time, data, required machine learning (ML) knowledge and estimated energy cost for each approach. Each method (listed in the first column) is paired with an approximate implementation time, the estimated dataset size, the level of ML expertise needed and estimated energy cost. These estimates assume that you have at least a bachelor's level of understanding of chemistry and at least some computational background. The energy costs are estimated by simulations[? ?] accounting for the number of model calls, architecture, model size and input length.

Model adaptation	Time	Data	ML knowledge	Energy Cost
Pre-training	Weeks	1M–1B+	Very High	50MWh - 1GWh
Zero-shot prompting	Minutes	None	None	<10Wh
Few-shot Prompting	Hours	<10	None	<1kWh
Fine-tuning	Days	<10k	High	100kWh - 50MWh
Coupling into systems				
RAG	Days	100k–1M+	Low	<1kWh
Tool-Augmentation	Days	None / 10k+	Low	<1kWh

Reviewer Point P 1.13 — In the introduction section, it says GPMs can "easily adapt to new settings" (Section 1), which I think is overly optimistic. While some foundational models can do a better job in transfer to knowledge, the word "easily" should probably be avoided here—it depends heavily on the task, the model, and the available data.

Reply: *We have now addressed this by rewriting the section, which now reads:*

> *A GPM is a model that has been pre-trained on a broad, heterogeneous corpus spanning multiple data modalities (text, images, graphs) or representations (e.g., common names, 3D coordinates, molecular images). It can be applied to a wide spectrum of downstream tasks that differ in objective (classification, regression, generation, reasoning), input format, and domain—ranging from natural-language processing to chemistry and vision—with little or no task-specific fine-tuning. A GPM supports zero-shot, few-shot, or transfer learning and can serve as the core component of autonomous agents.*

We also added Table 1 (see Table R1) to differentiate GPMs from other models:

Reviewer Point P 1.14 — There are a few exciting recent developments not mentioned, particularly those combining LLMs with knowledge graphs for scientific reasoning. Also, I've seen several examples in 2023–2025 of LLM+agent systems being used for real lab explorations. I'd also love to see more discussion to reasoning LLMs. ethero is mentioned. the architecture and implications of reasoning-focused models for chemical task will be interesting to readers.

Reply: *We have now improved the discussion about reasoning models by expanding our discussion of reasoning-focused architectures in section 5.2 (Existing GPMs for Chemical Science). We expanded the paragraph on Reasoning-Based Approaches, which now reads:*

> **REASONING-BASED APPROACHES** *A recent development in chemical GPMs incorporates explicit reasoning capabilities. The ethero model demonstrated this approach as a 24 billion-parameter reasoning model trained on over 640k chemically-verifiable problems (e.g., through code) across 375 tasks, including single-step retrosynthesis, molecular editing, and reaction prediction.[17] Unlike previous models, ethero's training used a novel RL approach. First, DeepSeek-R1 was prompted to create long chain-of-thought (CoT) traces ending in a SMILES. These traces were used to fine-tune a pre-trained Mistral-Small-24B using SFT, resulting in a "base reasoner" model. This model underwent RL using group-relative policy optimization (GRPO) on several verifiable chemical tasks to create "specialist" models—one for each task. High-quality outputs from these specialist models were selected and used to fine-tune the "base reasoner" to create a "generalist" model, capable of reasoning about all the tasks. In the last step, the generalist model undergoes another round of RL on the chemical tasks to improve the performance. This method shows that structured problem-solving approaches can significantly improve performance on complex chemical tasks without the need for massive domain-specific corpora.*

In the Post-Supervised Adaptation section (section 3.7), we added a worked-out example showing the plausibility of reasoning models for optimizing properties, aiming to motivate readers to explore reasoning-based models (shown below):

Optimizing solar cells

Goal: *Design perovskites that are both stable (high thermal/phase stability) and high-performing (optimal bandgap for light absorption).*
State: *Current partial perovskite formula (e.g., ABX_3 lattice with partial cation/anion substitutions).*

Action: *Substitute the next ion or additive (e.g., replace A-site with Cs^+ or add Cl^- dopant).*

Reward:

- `+1` *for stability (e.g., based on Goldschmidt tolerance factor).*
- `+1` *for bandgap tuning (*`1.5 to 1.8 eV` *ideal for single-junction cells).*
- `-1` *for defect proneness (e.g., halide migration).*

Episode Example:

- *Start with PbI_2 scaffold (B and X sites) → add MA^+ (methylammonium) at A-site → Reward:* `+0.5` *(good initial bandgap but low stability due to organic volatility).*
- *Co-substitute with Cs^+ and FA^+ (formamidinium) → Reward:* `+1.7` *(improved tolerance factor >*`0.9`*, stable mixed-cation perovskite with approx.* `1.6 eV` *bandgap).*
- *Over-dope with excess Br^- → Reward:* `-0.7` *(widens bandgap too much >*`1.8 eV`*, reducing PCE; also increases defects).*

The model learns to generate perovskite compositions that maximize cumulative reward, naturally discovering stable, high-PCE materials in the vast compositional space for next-gen solar cells. Note, however, that attempting this or a problem would require designing reward functions that can score actions taken by the agent.

However, as this is currently the only reasoning model specifically developed for the chemical sciences, the discussion is necessarily focused on this example.

Reviewer Point P 1.15 — Finally, I think the current manuscript title (while short and impressive) implies the field is already fully developed (especially for chemists who dont have prior knowledge in this area). A slight revision might help. Either modify title or abstract to let readers know this is an evolving topic. Otherwise, the title might suggest a kind of static completeness that doesn't reflect how fast the field is changing.

Reply: *We have updated the title and abstract to better reflect that this is an evolving field. The title now reads "General-Purpose Models for the Chemical Sciences: LLMs and Beyond". The abstract closes with*

> *"While many of these applications are still in the prototype phase, we expect that the increasing interest in GPMs will make many of them mature in the coming years."*

REVIEWER 2

Reviewer Point P 2.1 — The "General-Purpose Models for the Chemical Sciences" review by Alampara et al. is an extraordinarily ambitious summary (well over 170 pages) that seeks to chart the landscape of foundation-style AI models in chemistry. The authors have gathered an impressive amount of literature, most of which not related to the objective of the review.

Reply: *We thank the reviewer for acknowledging the ambition of our work. We have now extensively reviewed and pruned the references we cite, removing 50+ citations that were not directly relevant to GPMs. In particular, we also revised all sections in the application part to emphasize depth over breadth. For instance, in the section about property prediction, we now discuss fewer examples in the main text and moved many references to a comparative table.*

Reviewer Point P 2.2 — The manuscript is currently too dispersive, diffuse and uneven to meet Chemical Reviews standards. A sharper focus on genuine general-purpose (foundation) models, tighter organization, and a more consistently critical voice will be needed before the article can be recommended for publication in this venue. For these reasons, I recommend a major revision (borderline).

Reply: *We have now:*

- *Reviewed Sections 5, 6, and 7, making them more focused, discussing only works that add significant value to the topic.*
- *Provided a more critical voice by adding explicit "Limitations" and "Open Challenges" subsections for all applications discussed in Sections 5 and 6.*
- *We also now integrated "example boxes" throughout Section 3, which walks through key concepts with chemistry-specific examples. (See our response Point P 1.3)*
- *Sharpened our focus on genuine GPMs by adding a clear definition and comparison table (Table 1) early in the manuscript. (See our response to reviewer 1 Point P 1.2)*

Reviewer Point P 2.3 — The review introduces "general-purpose models" (GPMs), such as LLM, and then surveys literally everything from data formats and model architectures to molecule design, retrosynthesis, education, and ethics. That goes well beyond the objective stated in the abstract and missing the Chemical Reviews standards. The manuscript is overly ambitious in aiming to be a one-stop resource for how AI might transform every stage of chemical research, rather than focusing on GPMs.

Reply: *We have now reviewed the entire article, where we very early in the paper define and distinguish them from other types of models. See our response Point P 1.2, where we introduce a new table and definition.*

Apart from that, we have now extensively pruned different sections as mentioned in responses Point P 2.2 and Point P 2.1. For example, in the materials generation and property prediction section, we removed all the enumeration of incremental work and moved citations to a table (see Table R4). Apart from incremental works, we also pruned tangential discussions. For example, in experiment execution, we removed discussions related to protocol languages.

With the new changes, we now do not aim to review all existing approaches for a given application but rather show how GPMs have and could be used for various applications.

Reviewer Point P 2.4 — The stated focus is GPMs, yet much of the text drifts into robotics, classic QSAR, and other areas only tangentially related to foundation

Table R4: Non-comprehensive list of GPMs applied to property prediction tasks. The table presents different models and their applications across different molecular and materials property prediction benchmarks, showing the diversity of properties (from molecular toxicology to crystal band gaps), datasets used for evaluation, modeling approaches (prompting, fine-tuning, or retrieval-augmented generation), and task types (classification or regression.)

Model	Property	Dataset	Approach	Task
LLM-Prop[41]	Band Gap	CrystalFeatures-MP2022[41]	P	R
	Volume			R
	Band gap			C
LLM4SD[42]	Blood-brain barrier penetration	BBBP[43]	P	C
	FDA approval	ClinTox[44]		C
	Toxicology	Tox21[45]		C
	Drug-related side effects	SIDER[46]		C
	HIV replication inhibition	HIV[44]		C
	β-secretase binding	BACE[44]		C
	Solubility	ESOL[44]		R
	Hydration Free Energy	FreeSolv[47]		R
	Lipophilicity	Lipophilicity[44]		R
	Quantum Mechanics	QM9[44]		R
Domain Knowledge Prompt-Engineering[48]	Crystal	Custom[48]	P	C,R
	Organic Small Molecules	PubChem		
	Enzymes	UniProt		
MolecularGPT[49]	β-secretase binding	BACE[44]	P	C
	HIV replication inhibition	HIV[44]		C
	Bioactivity	MUV[50]		C
	Toxicology	Tox21[45]		C
	Toxicology	ToxCast[51]		C
	Blood-brain barrier penetration	BBBP[43]		C
	Cytochrome P450 isozymes	CYP450[52]		C
	Solubility	ESOL[53]		R
	Hydration Free Energy	FreeSolv[47]		R
	Lipophilicity	Lipophilicity[44]		R
GPT-MolBERTa[54]	Blood brain barrier penetration	BBBP[43]	P	C
	Toxicity	Tox21[45]		C
	Toxicity	Toxcast[51]		C
	FDA Approval	Clintox[44]		C
	Solubility	ESOL[53]		R
	Hydration Free Energy	FreeSolv[47]		R
	Lipophilicity	Lipophilicity[44]		R
	HIV replication inhibition	HIV[44]		C
	β-secretase binding	BACE[44]		C
GPT-Chem[55]	HOMO/LUMO	QMUGs[56]	FT	C, R
	Solubility	DLS-100[57]		C, R
	Lipophilicity	LipoData[55]		C, R
	Hydration Free Energy	FreeSolv[47]		C, R
	Photoconversion Efficiency	OPV[55]		C, R
	Toxicology	Tox21[45]		C, R
	CO_2 Henry Coefficients of MOFs	MOFSorb-H[58]		C, R
LLaMP[59]	Bulk modulus	Materials Project[60]	RAG	R
	Formation energy			R
	Electronic bandgap			R
	Multi-element electronic bandgap			R

Key: P = prompting; FT = fine-tuned model; RAG = retrieval-augmented generation; C = Classification; R = Regression models/GPMs. Removing these digressions would help readers see the wood for the trees. This would also serve in removing a large number of references that nothing have to do with GPMs, or even with AI (see a series of related to the Chemputer by Cronin - many other examples beyond this are present). Rather than a review on GPMs seems more a textbook on Digital Chemistry. Authors must perform some heavy clean-up and refocusing around the main objective of the work.

Reply:
As discussed in response to point P 2.3, we streamlined the discussion. In particular, we have made the experimental execution section more concise by removing references and discussions that diverged from GPMs. We removed approximately 7 references related to the Chemputer and similar topics (,[61],[62],[63],[64],[65],[66][67])

Reviewer Point P 2.5 — Several recent reviews and perspectives (e.g. https://doi.org/10.1039/D4SC03921A; https://doi.org/10.1021/jacsau.4c01160 and others) cover similar ground. Please articulate, early and explicitly, what new perspective this article provides. A distinctive taxonomy, unified framework, or meta-analysis of model performance could do the job.

Reply: *Our review provides several distinctive contributions that differentiate it from recent reviews:*

- *We introduce a clear taxonomy distinguishing GPMs from domain-specific foundation models and specialized chemistry pipelines (see Point P 1.2), with concrete examples and characteristic features for each category*
- *We provide a unified framework that integrates the entire scientific workflow—from data collection to hypothesis generation, experimental planning, execution, and reporting.*
- *We include critical "Limitations" and "Open Challenges" subsections for each application area, providing concrete failure cases and explicit discussion of when GPMs should and should not be used. Apart from this, we have a quantitative meta-analysis of model performance across different tasks, including comparative error distributions showing where LLMs underperform specialized models in property prediction tasks in the application section. (See for example Figure R3 below)*
- *We incorporated worked-out examples into our figures (see, for example, Figure R4)*
- *We provide substantially deeper technical coverage of fundamental building principles and design of evaluations (Sections 2-4) compared to existing reviews. Based on the recommendations from the reviewers, we have now also included chemistry-specific examples throughout the section. (see for example Example 6)*

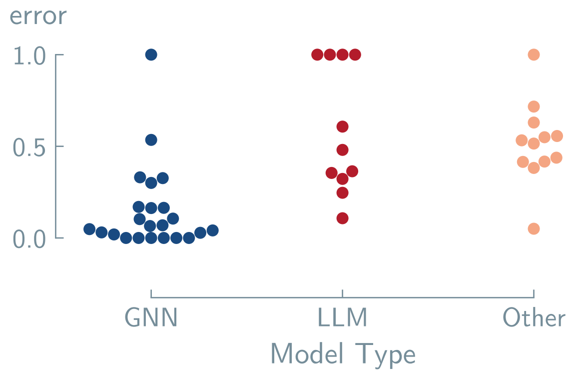


Figure R3: Normalized error distributions for materials property prediction models across different architectures. Each point represents the normalized error of a model on a specific property prediction task. Normalization was achieved with min/max values of each dataset to produce a range of errors between 0 and 1. The first column (blue) shows GNN based models, the second column (red) displays LLM approaches, and the third column (orange) represents other baseline methods and SOTA models, including *CrabNet*.[19] Lower values indicate better predictive performance.

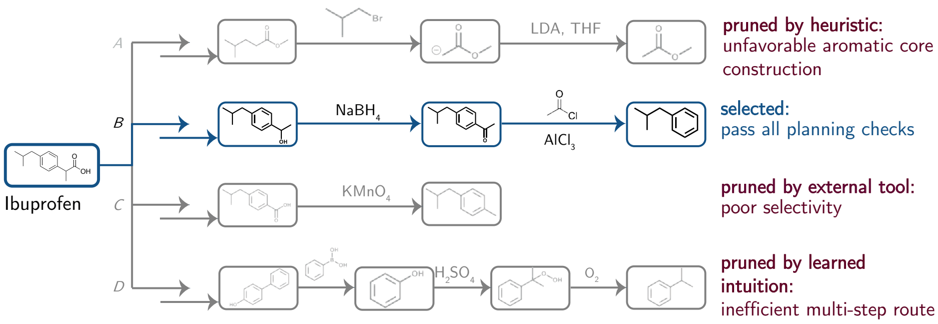


Figure R4: GPM-guided retrosynthesis route planning and pruning. GPMs can systematically evaluate and prune retrosynthetic routes using multiple reasoning capabilities to discriminate between viable and problematic approaches. The partially overlapping arrows at the start of each route indicate multiple steps. **Route A**: This route was pruned by heuristic reasoning due to the unfavorable aromatic core construction. **Route B**: This route was selected as it successfully passes all GPM planning checks, demonstrating optimal synthetic feasibility. **Route C**: This pathway was pruned by external tools due to the poor regio-selectivity of the oxidation step. **Route D**: This route was pruned based on learned intuition, as it represents an inefficient multistep pathway; the route could just start with phenol instead of synthesizing it.

Multimodal reasoning for chemical identification

A chemist finds an unknown white powder in a food sample which is highly soluble in water and tastes sweet. Chemist wants to identify it:
Input Modalities:

- ***^{13}C-NMR spectrum:*** *Shows peaks in the 60–100 ppm region, characteristic of carbons bonded to oxygen.*
- ***^{1}H-NMR spectrum:*** *Multiple peaks in the 3–5 ppm region with complex splitting patterns, indicating protons on oxygen-bearing carbons.*
- ***IR spectrum:*** *Broad, intense peak at 3000–3500 cm^{-1} (O-H stretching), peaks around 2900 cm^{-1} (C-H stretching), strong peaks at 1000–1150 cm^{-1} (C-O stretching, ether and alcohol).*
- ***Mass spectrometry:*** *Molecular ion peak at* `m/z = 180`.
- ***Text description:*** *"White crystalline solid, sweet taste, highly soluble in water, found in food sample".*

Multimodal Model Reasoning:

- ***MS + IR*** *→ Molecular formula $C_6H_{12}O_6$ with multiple hydroxyl groups (polyol).*
- ***NMR*** *→ Chemical shifts and splitting patterns consistent with a cyclic structure containing multiple oxygen-bearing carbons; presence of anomeric signals suggests equilibrium between ring forms.*
- ***Text + Chemical Knowledge*** *→ Sweet taste and high water solubility point to a simple sugar (monosaccharide).*

Based on the evidence, the model hypothesizes the molecule to be D-glucose. Single modality would be insufficient—mass spectrometry alone could match any hexose isomer (fructose, galactose, etc.), but combining spectroscopic fingerprints with physical properties enables confident identification.

We now write at the end of introduction,

> *We hope to contribute to this by addressing chemists and computer scientists, by providing technical background, a consistent terminology, and explaining key technical terminology in a glossary at the end of the manuscript.*

Reviewer Point P 2.6 — Many subsections read like annotated bibliographies. I would rather see fewer examples examined in greater depth (why they succeed, where they fail, and what they teach us) than exhaustive tallies of every preprint. Authors need to address this point.

Reply: *We have addressed this concern by substantially editing the applications sections (Sections 5-7). See our response to Point P 2.3*

Reviewer Point P 2.7 — The tone often feels boosterish. Each technical section should balance accomplishments with known limitations (hallucination rates, failure modes in retrosynthesis, domain-shift issues, data leakage, etc). The more candid you are, the more valuable the review becomes. Authors need to address this point.

Reply: *We have made substantial changes to provide a more balanced and candid tone. Specifically:*

We added explicit "Limitations" subsections in each application section discussing known failure modes, hallucination issues, domain-shift problems, data leakage concerns, and other critical limitations. See, for example, an excerpt from the GPMs as optimizers section on limitations and open challenges:

> **LIMITATIONS** *Current LLMs for chemical optimization, to date, exhibit a pronounced volatility, rarely yielding stable performance gains. Their outcomes are highly sensitive to prompt phrasing, and a lack of calibrated uncertainty estimates prevents their use for principled acquisition strategies.[68] Furthermore, these models frequently produce hallucinations and violate critical constraints, leading to invalid chemical structures or the proposal of unsafe and infeasible experimental conditions. Alleged improvements often hinge on narrow benchmarks, extensive pre- and post-processing, or access to downstream oracles, rather than on robust chemical reasoning. Within hybrid pipelines, the specific contribution of the LLM becomes difficult to isolate.*
>
> **OPEN CHALLENGES**
>
> - ***Correct Prompt Sensitivity***: *A key challenge is achieving prompt invariance, where the ranking of candidates remains stable under controlled rephrasings. Promising research directions include developing rigorous invariance tests and moving beyond discrete tokens to leverage models' more stable hidden states.[69]*
> - ***More robust evaluations***: *Future work should develop more robust frameworks that assess oracle realism, conduct thorough leakage audits to detect data contamination, and integrate uncertainty-aware metrics to better quantify risk and reliability.*
> - ***Ablations for the specific LLM role***: *There is a critical need for systematic ablation studies that isolate core LLM performance from enhancements like retrieval-augmented generation (RAG), external scorers, or tool use. Such studies are essential for fairly comparing general and fine-tuned LLMs and for understanding the source of performance gains.*

Reviewer Point P 2.8 — Self-citations dominate several sections. Cite your own work where it is genuinely seminal, of course, but be sure to give equal weight to other leading groups. This is a critical requirement of Chemical Reviews. Also reconsider whether every arXiv preprint needs to be included. The current selection of works is unreasonable and most important, unjustified by the objectives of the review.

Reply: *We have carefully reviewed our citations and made the following changes:*

- *Reduced citations to only those that are directly relevant and provide important contributions (removed more than 50+ citations overall).*
- *Applied stricter criteria for inclusion, asking whether each cited work adds substantive value to understanding GPMs in chemistry. We moved many incremental contributions from the main text to summary tables or removed them entirely if they did not add significant insights. (See Table R4, Table 8 and the property prediction section)*

Reviewer Point P 2.9 — Define "GPM" crisply and stick to it. Make clear whether it is synonymous with "foundation model," whether multimodal models qualify, and so on. Harmonise acronyms across sections.

Reply: *We have now added a crisp definition of GPM in Section 1 (Introduction) and stick to it throughout the manuscript:*

> *A GPM is a model that has been pre-trained on a broad, heterogeneous corpus spanning multiple data modalities (text, images, graphs) or representations (e.g., common names, 3D coordinates, molecular images). It can be applied to a wide spectrum of downstream tasks that differ in objective (classification, regression, generation, reasoning), input format, and domain—ranging from natural-language processing to chemistry and vision—with little or no task-specific fine-tuning. A GPM supports zero-shot, few-shot, or transfer learning and can serve as the core component of autonomous agents.*

We also added Table 1 (see Table R1) to clarify the distinction between GPMs, domain-specific foundation models, and specialized chemistry pipelines, with representative examples for each category. We explicitly explain that we adopt the term GPM to avoid semantic overlap with "foundation model" and to signal the breadth of applicability. Multimodal models are included as a subcategory of GPMs. We have harmonized acronym usage throughout all sections to maintain consistency.

Reviewer Point P 2.10 — Remove informal flourishes (e.g. the heart icon in the author list) and proof-read carefully, working on the entire manuscript with highest level of professionalism. There are encoding glitches ("flexibilitý") and scattered typos.

Reply: *We have now removed informal flourishes, including the heart icon in the author list, and reviewed the entire manuscript. We have specifically:*

- *Removed all non-standard icons and informal elements.*
- *Carefully proofread the entire manuscript, ensuring consistent formatting throughout.*

Reviewer Point P 2.11 — It could be even better if it proposed a short list of concrete, open research problems. In its current form, the review (through this section) does not address insights about the remaining challenges and future directions for the field. This is a critically important component of reviews that the author have to address substantially better.

Reply: *This feedback led us to add "Open Challenges" subsections in each of the application sections (Sections 5 and 6) and the evaluation best practices box. For example, see the open challenges in the experiment execution section and the molecular and material generation section,*

> - ***Grounding Natural Language to Laboratory Actions.*** *Translating ambiguous natural-language instructions ("heat gently") into precise, safe operations requires developing validation layers that detect physically impossible or hazardous actions before hardware execution.*
> - ***Universal Protocol Standards.*** *No widely adopted formalization standard exists. While languages like chemical description language (χDL) show promise, achieving interoperability across platforms requires community consensus on abstraction levels and device interfaces. s (MCPs) offer a potential path forward by enforcing consistent interfaces between GPMs, instruments, and verification layers.*
> - ***Autonomous Error Recovery.*** *Current systems cannot autonomously diagnose and recover from experimental failures. Developing general-purpose failure detection mechanisms and recovery strategies would enable truly autonomous operation.*
> - ***Multimodal Integration.*** *Chemists use diverse data types—spectra, chromatograms, TLC plates, and microscopy images. Integrating these modalities into GPM decision-making loops remains technically challenging but essential for human-level experimental reasoning.*

- ***Verification and Provenance.*** *Industrial and clinical applications require complete experimental provenance: every decision logged with reasoning traces, all parameters recorded, and outcomes traceable to specific model versions.*
- ***Robust Conditional Generation*** *Developing models that can reliably generate novel diverse structures that satisfy complex, multi-objective constraints (e.g., high stability, specific electronic properties, and synthesizability) without over-relying on retrieved training data examples.*
- ***Bridging the Synthesis Gap*** *Creating new validation metrics that better predict the synthetic feasibility of generated materials and molecules, moving beyond structural similarity toward realistic pathway prediction*
- ***Unified Multi-Scale Generation*** *Developing frameworks that are capable of simultaneously modeling different representations (e.g., text, graphs, 3D structures) and scales (e.g., from non-planar molecules to crystalline materials) within a single generative process*

Similarly, we added explicit "Limitations" subsections pointing out more fundamental open questions.

Reviewer Point P 2.12 — Most schematic figures are dense. Simplify captions and ensure that all symbols are defined.

Reply: We have revised the figures to improve clarity by simplifying figure captions to be more concise and focused, adding a glossary of symbols used in figures, reviewing figures to remove content that did not add value (e.g., removed Figure 23 from Section 7.1), and ensuring all symbols and notation are clearly defined either in the figure itself or in the caption.

Reviewer Point P 2.13 — Check every entry against ACS format and weed out duplicates or uncited items.

Reply: We have now checked all reference entries against ACS format requirements, removed duplicate entries and uncited items from the bibliography, and ensured all citations include complete information (authors, titles, journal names, years, volumes, and page numbers).

Reviewer Point P 2.14 — Consider whether ML background material that is not chemistry-specific can be shortened or moved online. This is a review, not a textbook for teaching classes.

Reply: One distinguishing feature of our review is that we attempt to provide a unified perspective that highlights that the principles underlying LLMs can also be applied to other model architectures and datasets. We made this clearer with revisions to the introduction. To make it even more accessible for a chemistry audience, we added chemistry-specific examples (see Point P 1.3).

REVIEWER 3

Reviewer Point P 3.1 — I thoroughly appreciated this comprehensive review by Jablonka and coworkers. While there are of course many recent articles surveying the effect of language models in chemistry, this goes a bit beyond those in scope; moreover, it does I think a particularly good job of discussing the requirements/procedures for data curation, standardization, tokenization, etc across domains, which are mostly not covered in articles employing or discussing the impact of models such as transformers. I commend the authors on their hard work to put this together.

Reply: *We greatly appreciate the reviewer's recognition of the value of our work in providing a rigorous review of GPMs in chemistry.*

EDITORIAL REQUESTS

Editorial Request 1 — Tables should be numbered consecutively with Arabic numerals. Footnote references should be in superscript italicized letters within the table and the footnotes should appear directly below the table in order.

Reply: We have reviewed all tables in the manuscript and ensured they are numbered consecutively with Arabic numerals. All table footnote references are now in superscript italicized letters, and footnotes appear directly below each table in the correct order.

Editorial Request 2 — All graphics should be cited within the manuscript text. Please note that you are missing in-text citations for the following graphics: Figures 8–11, 16, Table 2. For multipart graphics, authors do not need to cite all parts as long as one part or the whole is cited, e.g., citation of Fig.1a or Fig. 1 is okay.

Reply: We have added in-text citations for all previously uncited graphics, including Figures 8–11, 16, and Table 2. We have verified that all figures and tables in the manuscript are now properly cited in the text.

Editorial Request 3 — References must adhere to ACS journal format: include the list of authors, article titles, journal name, publishing year, volume and at least first page; for example: (1.) Hammes-Schiffer, S.; Soudackov, A. V. Proton-Coupled Electron Transfer in Solution, Proteins, and Electrochemistry. J. Phys. Chem. B 2008, 112, 14108-14123

Reply: We have reviewed all references and ensured they adhere to ACS journal format. All journal references now include the complete list of authors, article titles, journal names, publishing years, volumes, and at least the first page number.

Editorial Request 4 — Incomplete references:

- Please include at least the first page for each reference citation for the following incomplete journal references: 8, 12, 18, etc.
- Please include date of access for the following incomplete Website reference(s): 35, 56, 514, etc.
- Please include author names, title of article, name of repository (section name, if provided), submission date, DOI or URL (accessed YYYY-MM-DD) for the following incomplete preprint references: 21, 22 and 24, etc.
- Please include publisher for the following incomplete conferences/reports references: 67, 225, 261, etc.

Reply: *We have now:*

- *Added at least the first page number for all journal references (refs 8, 12, 18, and all others that were incomplete)*
- *Added dates of access in the format (accessed YYYY-MM-DD) for all website references (refs 35, 56, 514, and all others)*
- *Completed all preprint references with author names, article titles, repository names, submission dates, and DOI or URL with access dates (refs 21, 22, 24, and all others)*

- *Added publisher information for all conference and report references (refs 67, 225, 261, and all others)*

Editorial Request 5 — A TOC graphic is required and should fit in an area no larger than 3.25 in. × 1.75 in. (approx. 8.25 cm × 4.45 cm) and should have adequate resolution and clarity. Confirm that all text is legible at this size.

Reply: We have prepared a TOC graphic.

REFERENCES

(1) Jumper, J. et al. *Nature* **2021**, *596*, 583–589.

(2) Lin, Z.; Akin, H.; Rao, R.; Hie, B.; Zhu, Z.; Lu, W.; Smetanin, N.; dos Santos Costa, A.; Fazel-Zarandi, M.; Sercu, T.; Candido, S., et al. *bioRxiv* **2022**.

(3) Batatia, I.; Benner, P.; Chiang, Y.; Elena, A. M.; Kovács, D. P.; Riebesell, J.; Advincula, X. R.; Asta, M.; Avaylon, M.; Baldwin, W. J., et al. *arXiv* **2023**, DOI: `10.48550/arXiv.2401.00096`.

(4) OpenAI et al. *arXiv* **2023**, DOI: `10.48550/arXiv.2303.08774`.

(5) Grattafiori, A. et al. *arXiv* **2024**, DOI: `10.48550/arXiv.2407.21783`.

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